Proceedings of the Third International Conference on Automation, Computing and Renewable Systems (ICACRS-2024)
IEEE Xplore Part Number: CFP24CB5-ART; ISBN: 979-8-3315-3242-0

2024 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS) | 979-8-3315-3242-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICACRS62842.2024.10841754

AI-Powered Crop Suggestion, Yield Prediction, Disease Detection, and Soil Monitoring

N.Santha Raju
Asst Professor
Computer Science & Engineering (AIML & CS)
Godavari Global University,
Rajahmundry, A.P
santharaju@giet.ac.in

R.Tamilkodi
Professor
Computer Science & Engineering (AIML & CS)
Godavari Global University,
Rajahmundry, A.P
tamil@giet.ac.in

Vanaparthi Chandra Shekar
UG student
Computer Science & Engineering (CS)
Godavari Institute of Engineering & Technology Rajahmundry, A.P
chandrasekharvanaparthi@gmail.com

Bogadi Jaya Bharathi
UG student
Computer Science & Engineering (CS)
Godavari Institute of Engineering & Technology Rajahmundry, A.P
bogadijayabharathi@gmail.com

Koviri dinesh kumar
UG student
Computer Science & Engineering (CS)
Godavari Institute of Engineering & Technology Rajahmundry, A.P
koviridinesh1234@gmail.com

Yanamala sumanth
UG student
Computer Science & Engineering (CS)
Godavari Institute of Engineering & Technology Rajahmundry, A.P
sumanthyanamala8@gmail.com

ABSTRACT: India's agricultural sector is crucial for food security. However, factors such as climate change, soil quality, and pest infestations can significantly impact crop yields. This research aims to develop a machine learning-based model to predict crop yields accurately. The proposed model leverages advanced techniques like Inception ResNet V2, AlexNet, Random Forest, Decision Tree, KNN, Naive Bayes, Simple RNN, Logistic Regression, and Voting Classifier to analyze historical data and identify key factors influencing crop yield. By considering variables such as area, crop variety, and climatic conditions, the model provides valuable insights to farmers. The proposed model achieved a high accuracy of 91%, demonstrating its potential to significantly improve agricultural practices. By enabling farmers to make informed decisions regarding crop selection, planting, and resource allocation, this research contributes to sustainable agriculture and food security.

Keywords – Crop Suggestion, Crop Yield Prediction, Plant Disease Detection, Soil Moisture Monitoring Machine Learning Algorithms, Voting Classifier, Decision Tree, Inception-ResNet-v2, K-Nearest Neighbors, Naive Bayes

I. INTRODUCTION

Recently, agriculture and technology have converged to enhance food production and farming practices. Machine learning (ML) plays a crucial role in predicting agricultural yields and recommending fertilizers. Traditional crop management methods often fail to account for specific crop needs or varying agricultural conditions, whereas advanced ML algorithms can provide accurate, tailored suggestions by analyzing extensive datasets, including soil composition, weather patterns, and crop performance. Bhosale et al. [8] demonstrated the potential of a hybrid approach to assess agricultural data, showcasing how data analysis can forecast crop yields. By leveraging ML models, farmers can increase productivity while minimizing resource use and environmental impact. These algorithms can recommend the optimal type and amount of fertilizer, helping avoid overuse or underuse. They can also adapt to changing conditions, offering real-time advice to enhance crop yields and promote sustainable farming. Elavarasan and Vincent [9] highlight the role of deep reinforcement learning in fostering sustainable agricultural practices. The use of ML in crop management marks a significant advancement as the agriculture sector faces growing global demand and environmental sustainability challenges. This is particularly relevant in India, where agriculture contributes 15.4% of GDP and employs 31% of the workforce [10].

II. LITERATURE REVIEW

Machine learning (ML) has gained significant attention in recent years, particularly in agricultural production prediction. By leveraging climate, soil, and historical data, ML models help scientists predict crop yields, thereby enhancing agricultural productivity. Nigam et al. [1] improved agricultural yield forecasting through ML algorithms, data preprocessing, and feature selection. On complex, variable datasets, Random Forest (RF) outperformed Decision Trees (DT) and Support Vector Machines (SVM). Priya et al. [2] highlighted that ensemble methods, such as RF, improved prediction accuracy by combining model strengths. Mishra et al. [3] found that neural networks excelled at predicting complex nonlinear relationships, indicating that hybrid models could enhance agricultural yield projections. Shruti Kulkarni et al. discovered that complex neural networks outperformed regression models, though they required more computational power [4]. The CRY model by Ananthara et al. [5] reduces dataset noise using colony clustering, improving prediction accuracy. Awan and Sap [6] showed that kernel-based SVM models can handle nonlinear agricultural datasets effectively. Additionally, using fuzzy logic and time-series models, Bang et al. [7] improved yield forecasts under uncertain conditions, demonstrating the potential of hybrid agricultural forecasting models.

Data mining and ML have proven valuable for agricultural yield prediction. Gandhi and Armstrong [11]

Authorized licensed use limited to: RL Jalappa Institute of Technology. Downloaded on February 18,2026 at 07:06:36 UTC from IEEE Xplore. Restrictions apply.

found data mining useful for predicting rice crop production in tropical India. Bang et al. [12] used fuzzy logic to forecast agricultural yields based on temperature and precipitation with hybrid statistical ML models. Bhosale et al. [13] demonstrated that data analysis enhances agricultural yield forecasting using multiple data sources. Data mining for agricultural management improvement was also supported by Mishra et al. [14]. Suresh et al. [15] assessed important crop yields in Tamil Nadu using enhanced KNN computations and K-means clustering. Finally, Islam et al. [16] applied Deep Neural Networks (DNN) for crop selection and yield prediction in Bangladesh, showing the broader applicability of these techniques.

III. METHODOLOGY

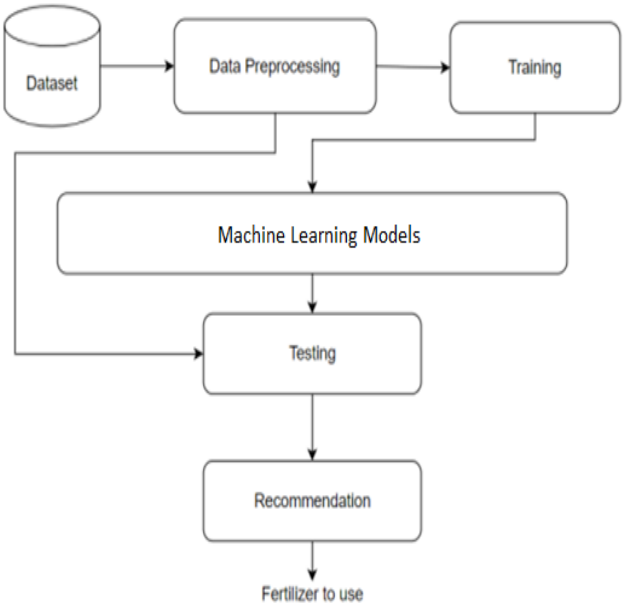


Figure 1 Proposed System Architecture

This research study develops a machine learning-based framework as shown in figure 1 for crop recommendation, yield prediction, plant disease diagnosis, and soil moisture monitoring. The framework utilizes various environmental factors like nitrogen, phosphorus, potassium, temperature, moisture, pH, and precipitation to suggest optimal crops for specific areas. A yield prediction module forecasts agricultural outputs based on parameters like season, crop type, and region. The plant disease detection component allows users to upload images of diseased plants for diagnosis and treatment suggestions based on image analysis. The soil moisture monitoring system gathers real-time data on soil moisture, temperature, humidity, wind speed, and other environmental factors to aid in efficient crop management. The system architecture involves preprocessing datasets for machine learning model training and selecting the best models for fertilizer recommendations, enhancing fertilizer application based on data-driven insights. The user interface, built using Flask with scikit-learn and TensorFlow, ensures smooth integration of machine learning models, improving agricultural outcomes and resource efficiency [17]. Data processing includes data cleaning, normalization, and feature selection to optimize model performance. During model training, 80% of the dataset is used to train the model, and 20% is reserved for testing the model's performance. Evaluation metrics like recall, accuracy, precision, and F1-score are used to assess the model. The study employs algorithms such as Inception-ResNet-v2 for image classification, Random Forest for ensemble learning, Decision Trees for supervised learning, and K-Nearest Neighbors (KNN) for classification based on similarity. Additionally, Naïve Bayes and Linear Regression are applied for probabilistic learning and regression analysis, respectively. Ensemble methods like bagging and boosting, along with Voting Classifiers, are used to enhance the model's robustness and adaptability [18-20]. By combining predictions from multiple models, these ensemble techniques improve the accuracy of crop yield predictions and disease detection, ultimately benefiting crop management.

IV. EXPERIMENTAL RESULTS

i) Dataset

The crop production dataset consists of 246,092 samples with attributes such as state name, district name, crop year, season, crop type, area, and production, providing detailed insights into regional crop yield trends. The crop recommendation dataset, with 2,201 samples, includes key agronomic and environmental attributes like nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The target label indicates the recommended crop type based on these factors. Additionally, the dataset will feature labeled plant images corresponding to various disease classes, enabling the detection of plant disease. Each attribute in these datasets was carefully selected to capture the critical conditions influencing crop selection and yield.

ii) Output of the Proposed System

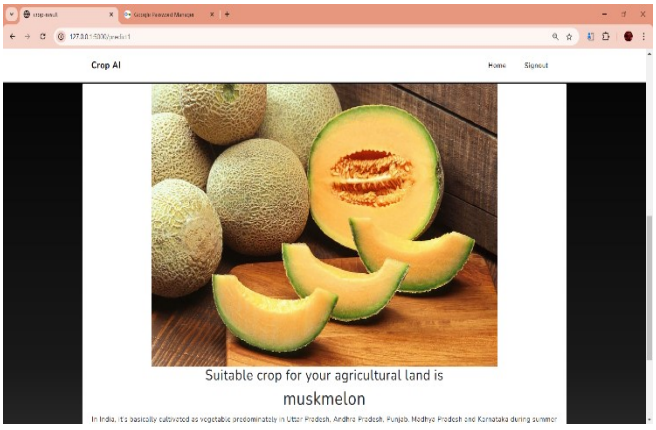


Figure 2 Resultant Crop Suggestion

Figure 2 shows the output of the crop suggested for planting. The prediction has been planned based on various algorithms used in the proposed method and the algorithm Which is shown in Table 1 shows the best one to pick for plantation.

Authorized licensed use limited to: RL Jalappa Institute of Technology. Downloaded on February 18,2026 at 07:06:36 UTC from IEEE Xplore. Restrictions apply.

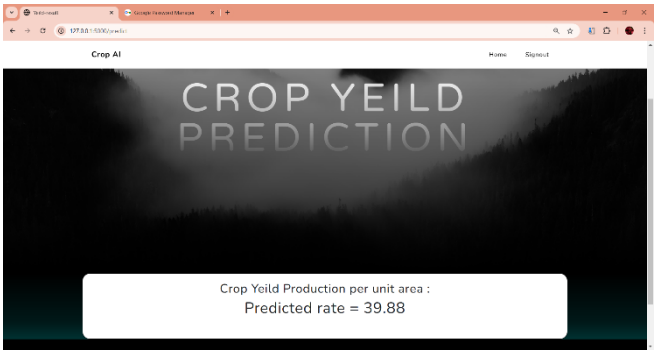


Figure 3 Resultant Crop Prediction Rate

Figure 3 shows the output of the crop's yield rate that is recommend by the proposed system. The best algorithm to predict the crop yields is indicated in Table 1 and has been utilized to plan the predictions based on a variety of algorithms employed in the proposed technique.

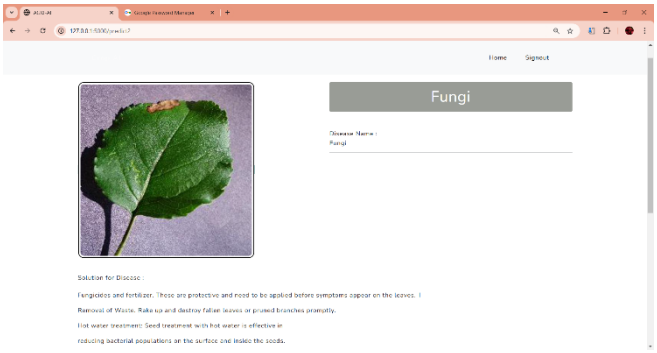


Figure 4 Plant disease detection result

Plant disease detection output is shown in Figure 4, which also serves as an illustration of the process. The model recognizes plant diseases using a variety of algorithms from the suggested approach. The method that offers the most precise detection for efficient plant health management is highlighted in Table 1.

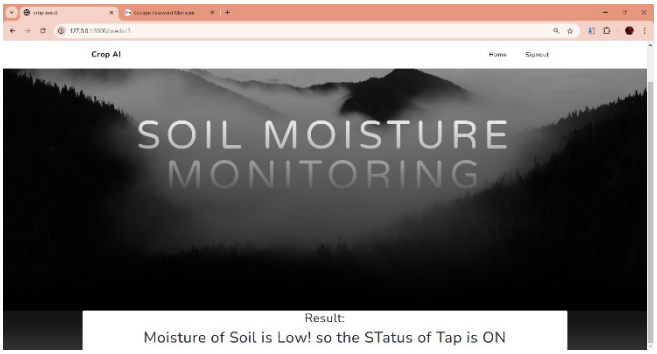


Figure 5 Final predicted soil moisture

Figure 5 shows the results of soil moisture monitoring after applying the proposed system methodology. Many algorithms are used to determine soil moisture content. Table 1 shows the most precise way to guarantee ideal irrigation and crop health care.

iii)Performance Evaluation:

Table 1 -Performance Evaluation Table

ML algorithm	Accuracy	Recall	Precision	F1 Score
Random Forest	0.99	0.99	0.99	0.99
Decision Tree	0.96	0.98	0.97	0.97
K-Nearest Neighbors	0.97	0.97	0.97	0.97
Navie Bayes	0.98	1.0	0.99	0.99

A comparative performance analysis of the DT, KNN, RF, and Navie Bayes algorithms is presented in Table 1 illustrating the accuracy, precision, recall, and F1 score that each algorithm achieved. The results of the proposed system were obtained from feature and data analysis carried out on the chosen dataset. The model's performance was evaluated using key metrics that were derived from the confusion matrix, including recall, accuracy, precision, and F1 score. The projected classifications are compared to the actual classifications in the dataset using a particular table structure called the confusion matrix. The following formulas were used to calculate the metrics:

Accuracy: Accuracy can be expressed as the ratio of correctly predicted cases, including true positives and true negatives, to all instances in the dataset.

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+FP+TN+FN)}$$

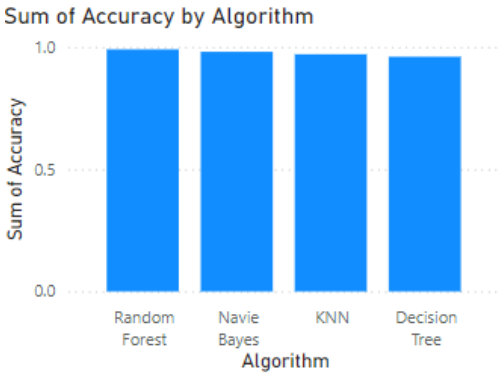


Figure 5 Representation of the Accuracy of the Proposed System

The proposed system's accuracy is demonstrated by the Figure 5, which displays high accuracy numbers for each algorithm. Table 1's data shows that Random Forest has the best accuracy (1.0), followed by Naive Bayes, KNN, and Decision Tree, all of which have excellent accuracy scores that are very near to 1.0.

Recall :

The ratio of true positives, or positively predicted cases, to all predicted positives, or true and false positives, is known as precision, sometimes known as the positive predictive value.

$$\text{Recall} = \frac{TP}{(TP+FN)}$$

Authorized licensed use limited to: RL Jalappa Institute of Technology. Downloaded on February 18,2026 at 07:06:36 UTC from IEEE Xplore. Restrictions apply.

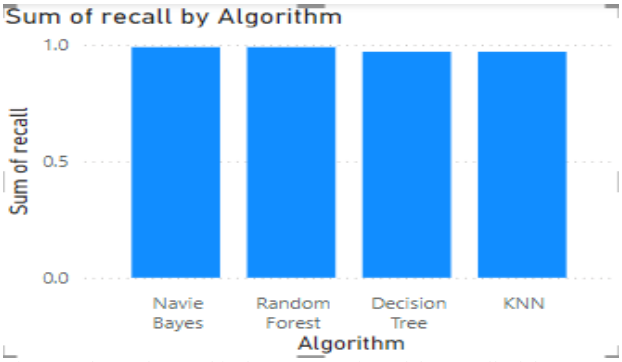


Figure 6 shows the graphical Representation of the Recall of the proposed System.

Figure 6 shows excellent recall values for all algorithms, demonstrates the recall of the proposed method. Table 1 data shows that NB has the highest recall (1.0), followed by DT, RF, and KNN, all of which have strong recall scores that are in the neighbourhood of 1.0.

F1-Score:

The percentage of genuine positives that were accurately identified from the real positive events is measured by recall.

$$\text{F1 Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

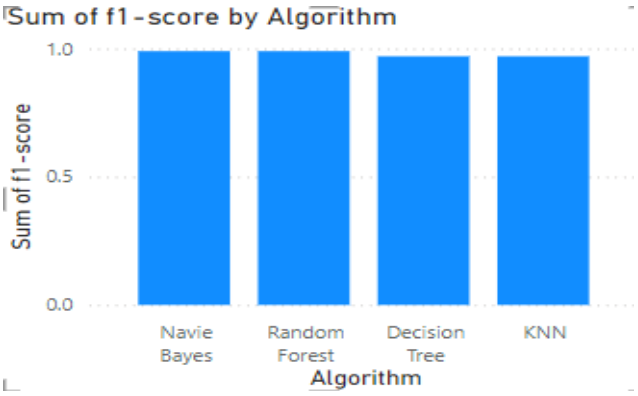


Figure 7 Graphical Representation of the F1-Score of the proposed System.

Figure 7 displays the F1-score of the proposed system and demonstrates high F1-score values for all algorithms. Naive Bayes and Random Forest have the greatest F1-scores (0.99) among the algorithms based on Table 1 data; Decision Tree and KNN also show strong F1-scores, both near 0.97.

Precision : It offers a balance between the two, particularly in situations when you must give equal weight to false positives and false negatives.

$$\text{Precision} = \frac{TP}{(TP+FP)}$$

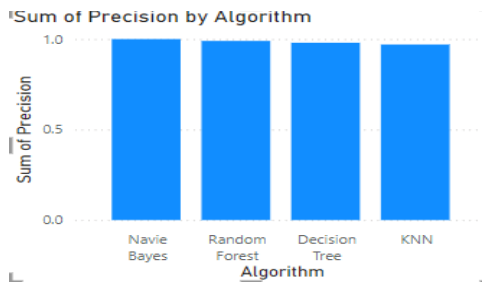


Figure 8 Graphical Representation of the Precision of the proposed System.

Figure 8 demonstrates high precision values for all algorithms, illustrates the precision of the suggested system. Table 1's data shows that Naive Bayes has the highest precision (1.0), followed by Random Forest, Decision Tree, and KNN, all of which have excellent precision scores that are very near to 1.0.

Table 2 – Comparison between Existing system and Proposed system

Reference	Model/System	Accuracy (%)	Features/Approach
Crop Prediction [1]	Logistic Regression	80%	Uses basic soil and climate data, limited crop diversity.
Crop Recommendation [2]	Decision Tree	85%	Simple decision tree based on soil type, crop growth patterns.
Disease Detection [3]	Random Forest	82%	Utilizes environmental data, lacks advanced disease markers.
Crop Prediction [4]	Neural Network (AlexNet)	88%	Neural network with feature extraction but limited dataset application.
[9] Disease Detection	InceptionNet	89%	CNN- image based disease recognition.
Proposed	Voting Classifier (Inception + ResNet V2 + AlexNet)	91%	Ensemble approach combining CNNs, trained on comprehensive data, including soil properties, crop types, and environmental factors.

Table 2 compares the accuracy and methodologies of various studies on crop prediction, recommendation, and disease detection, highlighting the effectiveness of machine learning techniques in enhancing agricultural practice.

V. CONCLUSION

In conclusion, this system utilizes advanced machine learning techniques to address critical aspects of modern agriculture, including crop recommendation, yield prediction, plant disease identification, and soil moisture monitoring. By analyzing essential parameters like soil nutrients, weather conditions, and environmental factors, the system offers precise crop suggestions and yield forecasts. The integration of image-based plant disease detection and real-time soil moisture monitoring further enhances the platform's utility, providing farmers with timely insights and actionable recommendationsThis all-encompassing strategy seeks to enhance agricultural output, maximize resource utilization, and encourage sustainable farming methods. Ultimately, the proposed system possesses the ability to empower farmers with data-driven tools to make informed decisions, enhancing crop health and maximizing yields.

Authorized licensed use limited to: RL Jalappa Institute of Technology. Downloaded on February 18,2026 at 07:06:36 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] A. Nigam, S. Garg, A. Agrawal and P. Agrawal, "Crop Yield Prediction Using Machine Learning Algorithms," 2019 Fifth International Conference on Image Information Processing (ICIIP), Shimla, India, 2019

[2] Priya, P., Muthaiah, U., Balamurugan, M.Predicting Yield of the Crop Using Machine Learning Algorithm,, Volume 9 Issue 4, April 2020 http://www.ijsr.net2015 www.ijsr.net2015

[3] Mishra, S., Mishra, D., Santra, G. H.,Applications of machine learning techniques in agricultural crop production, Indian Journal of Science and Technology 10.17485/ijst/2016/v9i38/95032 volume 9,2016

[4] Shruti Kulkarni, Shahnawaz Mandal, G Srivatsa Sharma, "Predictive Analysis to Improve Crop Yield using a Neural Network Model" IEEE-2018

[5] Ananthara, M. G., Arunkumar, T., & Hemavathy, R. (2013, February). CRY—an improved crop yield prediction model using bee hive clustering approach for agricultural data sets. In 2013 International Conference on Pattern Recognition, Informatics and Mobile Engineering (pp. 473- 478). IEEE.

[6] Awan, A. M., & Sap, M. N. M. (2006, April). An intelligent system based on kernel methods for crop yield prediction. In Pacific-Asia Conference on Knowledge Discovery and Data Mining (pp. 841-846). Springer, Berlin, Heidelberg.

[7] Bang, S., Bishnoi, R., Chauhan, A. S., Dixit, A. K., & Chawla, I. (2019, August). Fuzzy Logic based Crop Yield Prediction using Temperature and Rainfall parameters predicted through ARMA, SARIMA, and ARMAX models. In 2019 Twelfth International Conference on Contemporary Computing (IC3) (pp. 1-6). IEEE.

[8] Bhosale, S. V., Thombare, R. A., Dhemey, P. G., & Chaudhari, A. N. (2018, August). Crop Yield Prediction Using Data Analytics and Hybrid Approach. In 2018 Fourth International Conference on Computing Communication Control and Automation (ICCUBEA) (pp. 1-5). IEEE.

[9] D. Elavarasan and P. M. D. Vincent, "Crop Yield Prediction Using Deep Reinforcement Learning Model for Sustainable Agrarian Applications", IEEE Access, vol. 8, pp. 86886-86901, 2020.

[10] S. Bhanumathi, M. Vineeth and N. Rohit, "Crop Yield Prediction and Efficient use of Fertilizers", 2019 International Conference on Communication and Signal Processing (ICCSP), pp. 0769-0773, 2019.

[11] N. Gandhi and L. J. Armstrong, "Rice crop yield forecasting of the tropical wet and dry climatic zone of India using data mining techniques", 2016 IEEE International Conference on Advances in Computer Applications (ICACA), pp. 357-363, 2016.

[12] S. Bang, R. Bishnoi, A. S. Chauhan, A. K. Dixit and I. Chawla, "Fuzzy Logic based Crop Yield Prediction using Temperature and Rainfall parameters predicted through ARMA SARIMA and ARMAX models", 2019 Twelfth International Conference on Contemporary Computing (IC3), pp. 1-6, 2019.

[13] S. V. Bhosale, R A. Thombare, P. G. Dhemey and A. N. Chaudhari, "Crop Yield Prediction Using Data Analytics and Hybrid Approach", 2018 Fourth International Conference on Computing Communication Control and Automation (ICCUBEA), pp. 1-5, 2018.

[14] S. Mishra, P. Paygude, S. Chaudhary and S. Idate, "Use of data mining in crop yield prediction", 2018 2nd International Conference on Inventive Systems and Control (ICISC), pp. 796-802, 2018.

[15] A. Suresh, P. Ganesh Kumar and M. Ramalatha, "Prediction of major crop yields of Tamilnadu using K-means and Modified KNN", 2018 3rd International Conference on Communication and Electronics Systems (ICCES), pp. 88-93, 2018.

[16] T. Islam, T. A. Chisty and A. Chakrabarty, "A Deep Neural Network Approach for Crop Selection and Yield Prediction in Bangladesh", 2018 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), pp. 1-6, 2018.

[17] S. Veenadhari, B. Misra and C. Singh, "Machine learning approach for forecasting crop yield based on climatic parameters", 2014 International Conference on Computer Communication and Informatics, pp. 1-5, 2014.

[18] P. S. Nishant, P. Sai Venkat, B. L. Avinash and B. Jabber, "Crop Yield Prediction based on Indian Agriculture using Machine Learning", 2020 International Conference for Emerging Technology (INCET), pp. 1-4, 2020.

[19] A. Nigam, S. Garg, A. Agrawal and P. Agrawal, "Crop Yield Prediction Using Machine Learning Algorithms", 2019 Fifth International Conference on Image Information Processing (ICIIP), pp. 125-130, 2019.

[20] N. Suresh et al., "Crop Yield Prediction Using Random Forest Algorithm", 2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS), pp. 279-282, 2021.

Authorized licensed use limited to: RL Jalappa Institute of Technology. Downloaded on February 18,2026 at 07:06:36 UTC from IEEE Xplore. Restrictions apply.